Procurement and contract fraud

Bid rigging, change orders and false set-asides

Status: Draft v0.1 (2026-11-04), for Steve's review. Extension 6 of 10 (see the extensions overview). Reference code: `analytics/forensic/procurement.py` and `analytics/forensic/bids.py`, tested on synthetic data only.

Bottom line

The government buys about $750 billion a year of goods and services through contracts, and every award is published in FPDS and USAspending. Three schemes leave traces in those records: **weak or rigged competition**, **contracts that grow after they are won**, and **small business set-asides won by firms that are not small or do not do the work**. This extension turns each into a screen that runs on public data across every agency, so DoD-1, VA-1, GSA-1 and DOT-1 share one engine.

1. Data

Source	What it gives	Notes
FPDS / USAspending contract transactions	Every action: base and current value, offers received, extent competed, set-aside type, NAICS, PSC, office, modifications with reason codes	Award-level and transaction-level downloads; fields named as in the USAspending data dictionary
SAM.gov entity data	Size representations by NAICS, socioeconomic certifications, parents, officers, addresses	Self-certified; size is measured at offer
SBA Dynamic Small Business Search, certification lists (8(a), HUBZone, WOSB, VetCert)	Certified status and dates	VA's SDVOSB verification moved to SBA in 2023
FFATA subaward reports	First-tier subcontracts on larger awards	Underreported; absence is itself a signal
SBA size and status protest decisions, OHA decisions	Firms found other than small	Labeled cases for validation
DOJ Procurement Collusion Strike Force and contractor fraud releases	Charged cases	Labeled cases
State DOT bid tabulations	All bids, not only winners	For sealed-bid screens (DOT-1)

2. Screens

2.1 Competition

- **Single-offer rate** by office and market: competed awards that drew one offer. Persistently high rates point to tailored requirements or a steered market. Compared within NAICS, since some markets are thin.
- **Incumbent lock-in:** Herfindahl index of vendor shares by office and NAICS; high concentration with mostly single-offer competitions.
- **Sealed-bid distribution screens and rotation tests** (handbook section 6): low coefficient of variation, high relative distance, winners taking turns.

2.2 Change-order and modification inflation

- **Growth ratio:** current value / base and all options value at award. Robust z-score of its log within NAICS; awards above 3.5 are reviewed.
- **Many modifications just under an approval threshold** (warrant limits, cost or pricing data thresholds, legal review levels): the modification version of a split purchase.
- **Out-of-scope growth:** modifications coded as "additional work (new agreement)" or "supplemental agreement" on sole-source bridges.

2.3 Set-aside eligibility

Check	Rule	Source
Size	Average annual receipts or employees above the NAICS size standard	13 CFR 121.201; SAM representations, public filings
Pass-through	Prime pays more than the limit to firms that are not similarly situated	FAR 52.219-14: 50% for services and supplies, 85% general construction, 75% specialty trade
Affiliation and ostensible subcontractor	Set-aside winner shares owners, officers, address or bank account with a large firm	13 CFR 121.103; network method
Front-company profile	Set-aside share of revenue jumps by more than 40 points while awards at least double	Award history
Status after award	Firm acquired by or merged into a large business but keeps receiving set-aside orders without recertifying	SAM, FPDS recertification fields

Each reason is a question for the contracting officer or SBA, not a finding. Size is measured on the date of the offer, and affiliation needs the facts behind a shared address.

3. Thresholds to know

Thresholds change with inflation adjustments (FAR 1.109). The October 2025 adjustment raised the micro-purchase threshold to $15,000 and the simplified acquisition threshold to $350,000; check

the current FAR 2.101 before relying on these. Bunching tests (handbook section 5.4) look for excess awards just under each threshold.

4. Validation

- Positive set: SBA size protests sustained, OHA decisions, DOJ set-aside fraud cases, suspension and debarment records with procurement causes.
- Release bar as in the review plan: known cases at least five times more common in the top 5% than overall.
- False positives to expect: legitimate scope growth on research and development, emergency buys, sole-source awards under statute, joint ventures that are allowed to share resources.

5. Where it runs first

Idea	Screens	Partner
DOD-1 set-aside fronts	Size, pass-through, affiliation	DoD OIG, DCIS, SBA OIG
DOD-2 price and growth outliers	Growth ratio, threshold bunching	DoD OIG, DCAA
VA-1 SDVOSB fronts	Affiliation, front profile	VA OIG
GSA-1 schedule and MAS orders	Single-offer, lock-in, growth	GSA OIG
DOT-1 highway bid rigging	Sealed-bid screens, rotation	DOT OIG, DOJ Antitrust Division

6. Next steps

1. Pull FY2021-FY2026 contract transactions for DoD, VA and GSA (public).
2. Build the growth and single-offer tables by office (week 6).
3. Back-test set-aside checks on sustained size protests (week 8).